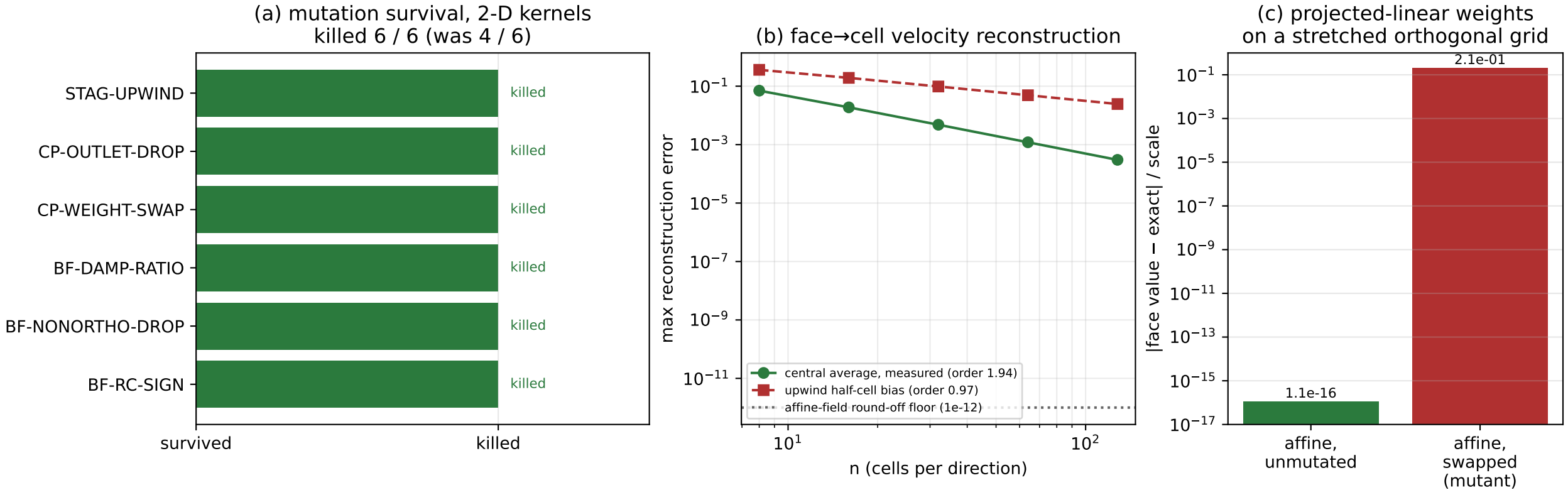
(a) mutation survival, 2-D kernels
killed 6 / 6 (was 4 / 6)
STAG-UPWIND
CP-OUTLET-DROP
CP-WEIGHT-SWAP
BF-DAMP-RATIO
BF-NONORTHO-DROP
BF-RC-SIGN
killed
survived
killed
(b) face→cell velocity reconstruction
max reconstruction error
n (cells per direction)
central average, measured (order 1.94)
upwind half-cell bias (order 0.97)
affine-field round-off floor (1e-12)
(c) projected-linear weights
on a stretched orthogonal grid
|face value − exact| / scale
2.1e-01
1.1e-16
affine,
unmutated
affine,
swapped
(mutant)